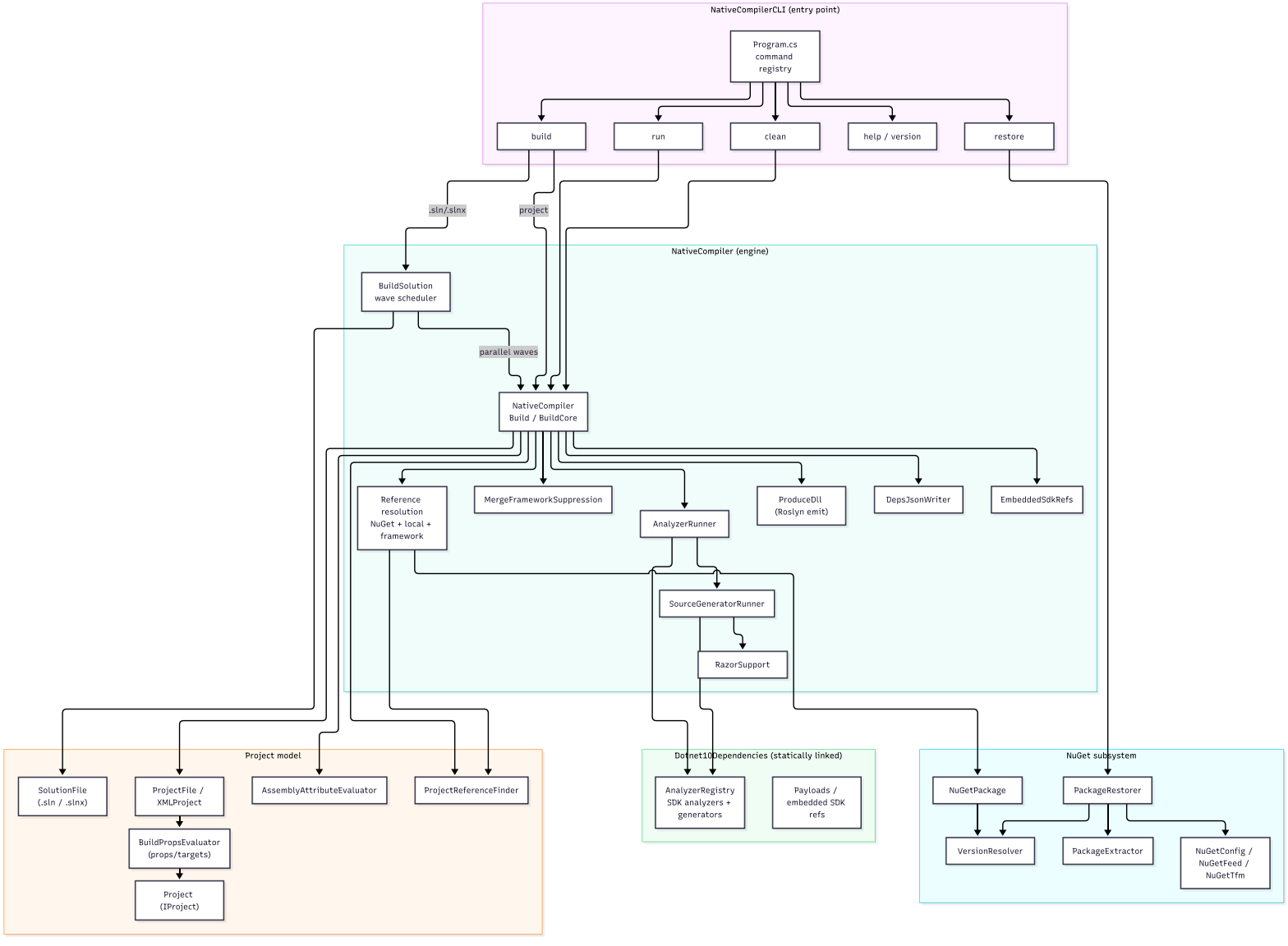
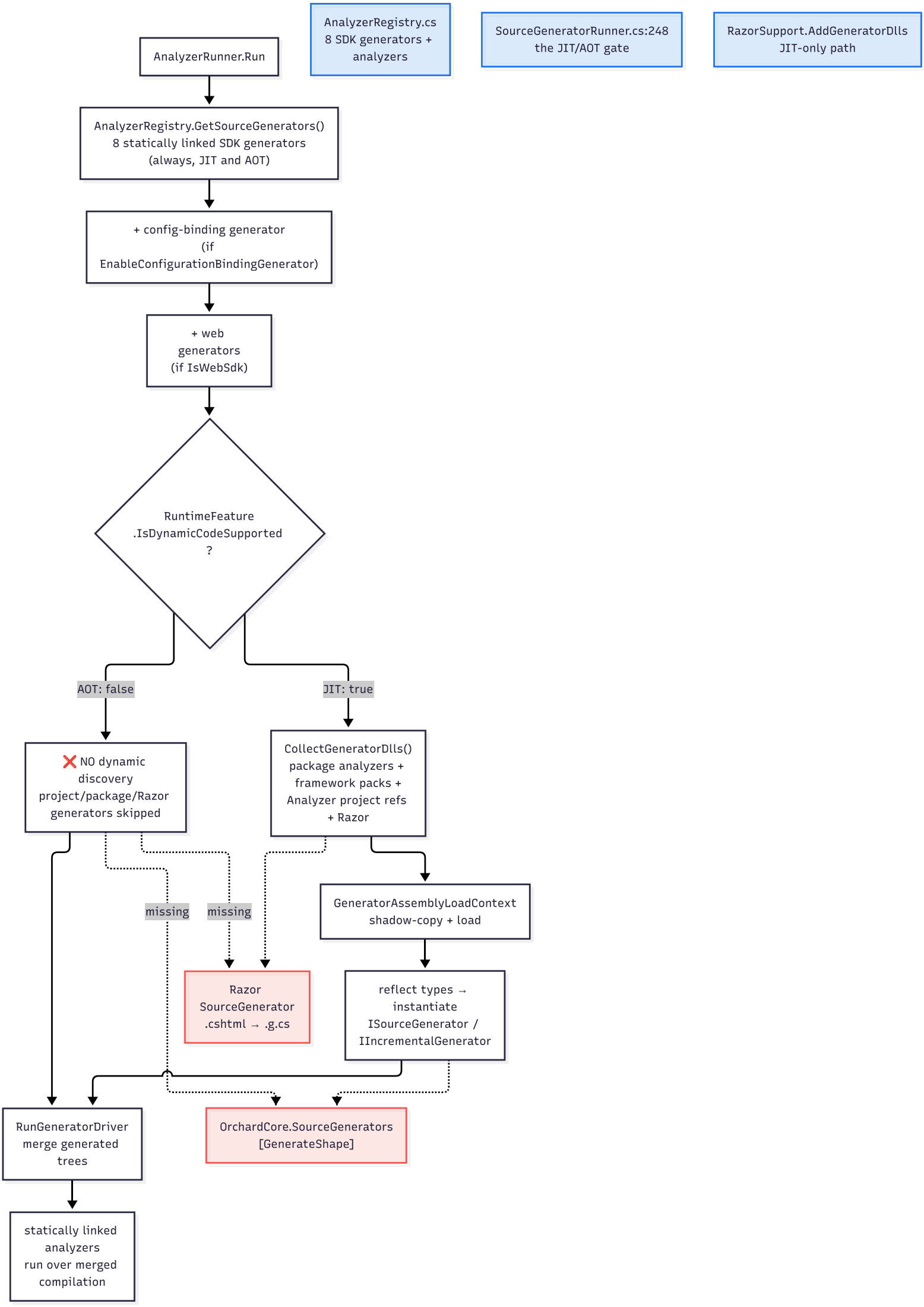
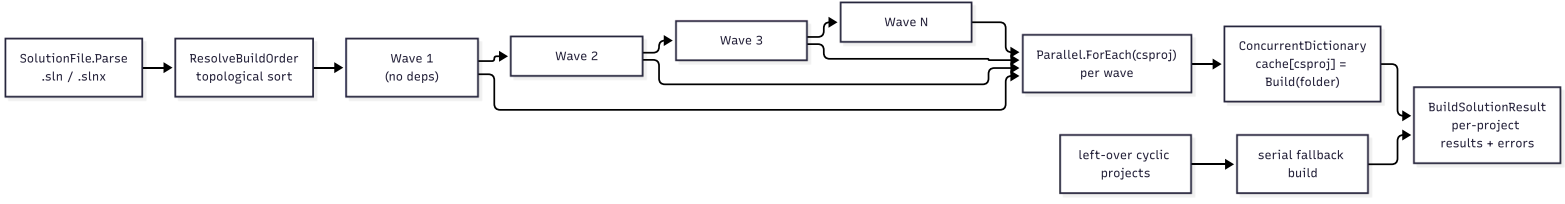
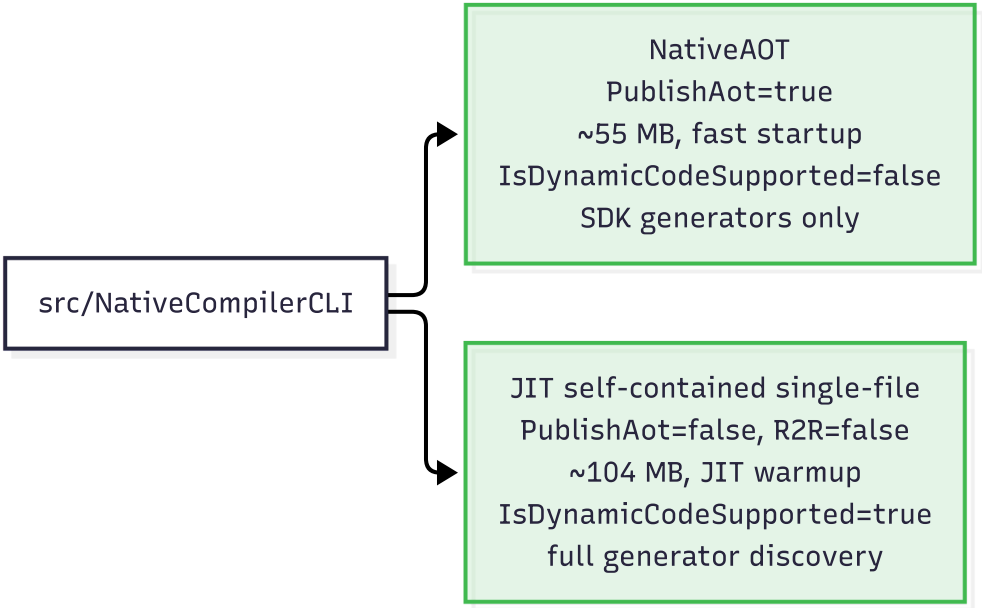

NativeCompilerCLI (entry point)
Program.cs command registry
build
run
clean
help / version
restore
.sln/.slnx
project
NativeCompiler (engine)
BuildSolution wave scheduler
parallel waves
NativeCompiler Build / BuildCore
Reference resolution NuGet + local + framework
MergeFrameworkSuppression
AnalyzerRunner
ProduceDll (Roslyn emit)
DepsJsonWriter
EmbeddedSdkRefs
SourceGeneratorRunner
RazorSupport
Project model
SolutionFile (.sln / .slnx)
ProjectFile / XMLProject
AssemblyAttributeEvaluator
ProjectReferenceFinder
BuildPropsEvaluator (props/targets)
Project (IProject)
Dotnet10Dependencies (statically linked)
AnalyzerRegistry SDK analyzers + generators
Payloads / embedded SDK refs
NuGet subsystem
NuGetPackage
PackageRestorer
VersionResolver
PackageExtractor
NuGetConfig / NuGetFeed / NuGetTfm

Build
projectPath
BuildsInProgress?
circular ref
yes
no
throw circular
reference
ReadAllProjectFilesPath
ReadAllFilesInMemory
GetProjectDetails
→ Project model
GetCompilationOptions
ResolveSdkRefLocation
embedded SDK refs
ParseProject
syntax trees +
embedded texts
GetdefaultLocalReferences
(ProjectReference DLLs)
GetdefaultNugetReferences
(package graph)
GetFrameworkReferenceCompilingRefs
MergeFrameworkSuppression
(dedup + keep HIGHEST
version)
CSharpCompilation.Create
AnalyzerRunner.Run
generator/analyzer
errors?
yes
no
BuildResult
failure
(errors, no emit)
needs test
entry point?
yes
no
inject entry-
point shim
ProduceDll
(Roslyn emit)
GetOutputRuntimeFiles
GetHintPathCopyLocalFiles
DepsJsonWriter.Write
WriteFiles →
sharp bin
GetOutputNativeFiles
(native assets)
references
NativeCompiler?
yes
no
EmbeddedSdkRefs.WritePayloadTo
BuildResult
success


AnalyzerRunner.Run
AnalyzerRegistry.cs
8 SDK generators + analyzers
SourceGeneratorRunner.cs:248
the JIT/AOT gate
RazorSupport.AddGeneratorDlls
JIT-only path
AnalyzerRegistry.GetSourceGenerators()
8 statically linked SDK generators
(always, JIT and AOT)
+ config-binding generator
(if EnableConfigurationBindingGenerator)
+ web generators
(if IsWebSdk)
RuntimeFeature .IsDynamicCodeSupported ?
AOT: false
JIT: true
❌ NO dynamic discovery
project/package/Razor generators skipped
CollectGeneratorDlls()
package analyzers + framework packs + Analyzer project refs + Razor
missing
missing
GeneratorAssemblyLoadContext
shadow-copy + load
Razor SourceGenerator
.cshtml → .g.cs
reflect types → instantiate
ISourceGenerator / IIncrementalGenerator
RunGeneratorDriver
merge generated trees
OrchardCore.SourceGenerators
[GenerateShape]
statically linked analyzers
run over merged compilation


SolutionFile.Parse
.sln / .slnx
ResolveBuildOrder
topological sort
Wave 1
(no deps)
Wave 2
Wave 3
Wave N
Parallel.ForEach(csproj)
per wave
ConcurrentDictionary
cache[csproj] =
Build(folder)
left-over cyclic
projects
serial fallback
build
BuildSolutionResult
per-project
results + errors


src/NativeCompilerCLI
NativeAOT
PublishAot=true
~55 MB, fast startup
IsDynamicCodeSupported=false
SDK generators only
JIT self-contained single-file
PublishAot=false, R2R=false
~104 MB, JIT warmup
IsDynamicCodeSupported=true
full generator discovery